Adaptive Perceptual Training via Simulation-Based Inference of GRT Models

Fitting a perceptual model fast enough to teach with it

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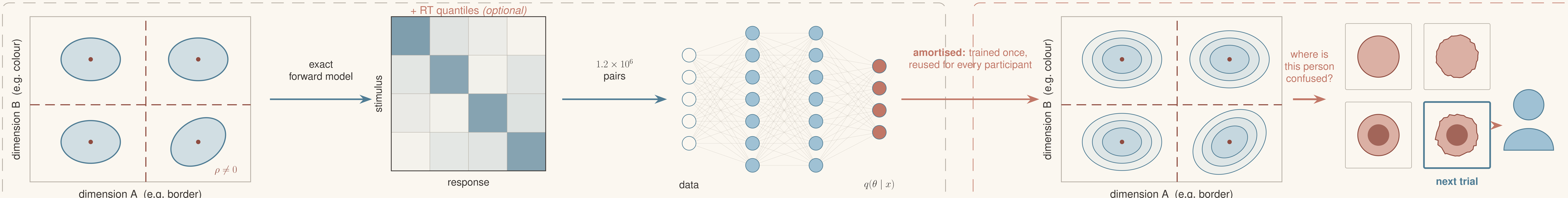
[osu-logo.png goes here]

code + design record

run it yourself

OFFLINE • done ONCE • hours on one GPU

ONLINE • EVERY TRIAL • microseconds



1 Sample a representation

Locations and noise of the four stimuli in perceptual space and where the decision bounds fall.

2 Simulate the data

Pushing θ through the exact forward model to produce matrices is cheap to generate **millions** of (θ, x) pairs.

3 Learn the inverse

Train a network to map data back to θ with a **full posterior**. Expensive, but paid **once**.

4 Infer, in microseconds

A real matrix in, a calibrated posterior out in ~3 µs. MLE needs ~90 ms and sometimes fails; this **always** answers.

5 Choose the next stimulus

Present the stimulus that targets the confusion we just measured — then re-infer, and go again.

every response updates the matrix, and the posterior moves with it

THE PROBLEM

What are we actually trying to measure?

Two trainees can score *identically* on a diagnostic task and be wrong in completely different ways. One cannot see the feature. One sees it but has drifted their criterion. One sees both features but cannot pull them apart.

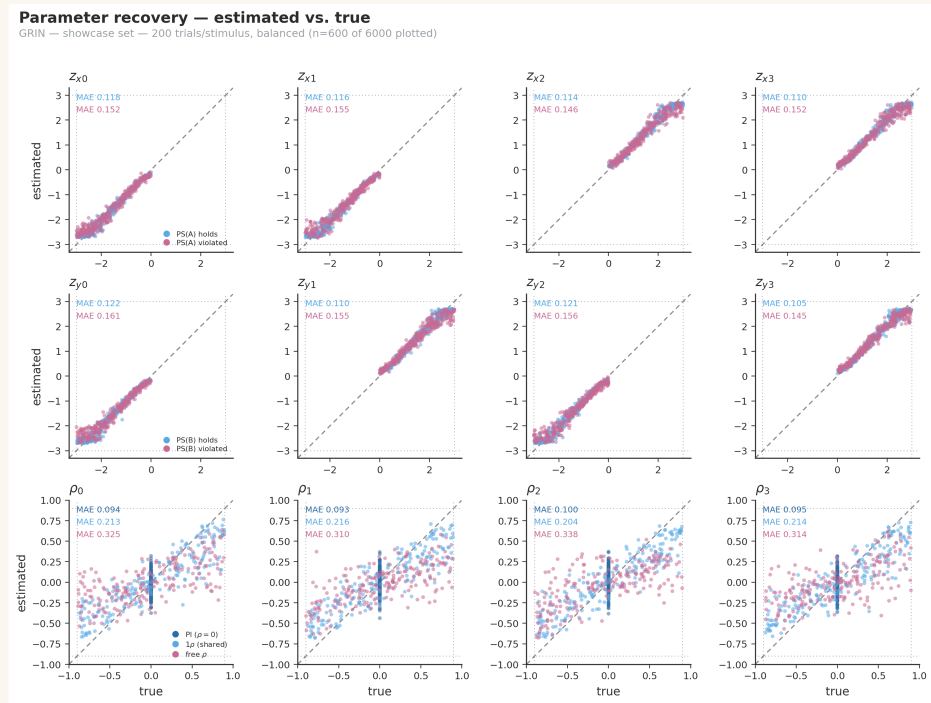
General Recognition Theory separates these. It gives each person a perceptual space: where the stimuli sit, how noisy each is, whether the two features are perceived independently, and where the decision bounds fall.

> **Why it matters clinically.** A trainee who confuses an irregular *border* with variegated *colour* needs different practice from one who simply cannot see colour variation at all. Accuracy alone cannot tell them apart. The representation can.

A 4×4 identification matrix carries exactly 12 independent numbers, so we work in the 12-parameter identified coordinates. Saturated, therefore identified — no restricted-model hierarchy needed.

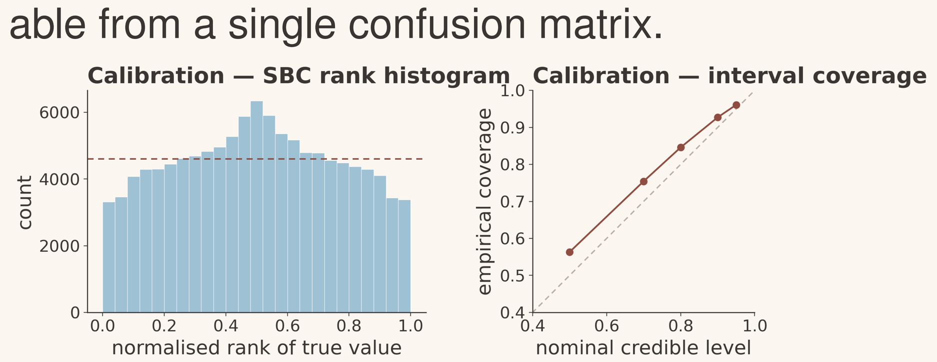
RECOVERY

Does it recover the truth?



Predicted vs. true on held-out simulations. Sensitivities are recovered tightly; correlations are not — and that is a property of the data, not the method.

Sensitivities come back at $r \approx 0.98$. Perceptual correlations are much noisier, $r \approx 0.75$. Maximum likelihood shows the *same* split on the same matrices: weak correlations are genuinely unrecoverable from a single confusion matrix.

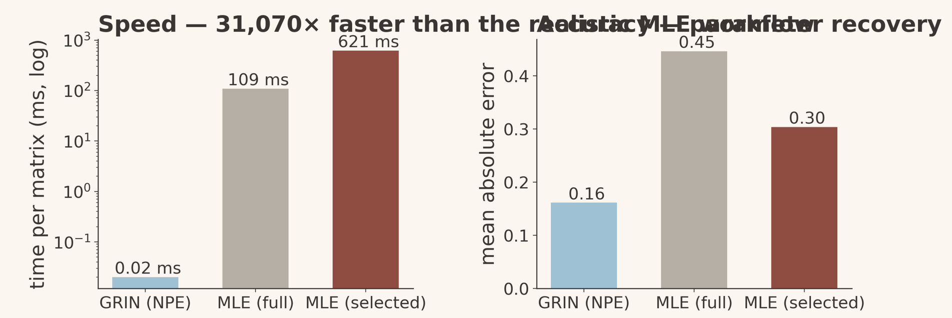


Simulation-based calibration: flat rank histogram, coverage on the diagonal.

The posterior is not just narrow — it is **honest**. That is what makes it safe to act on mid-experiment.

SPEED

Is it fast enough to matter?



Against the MLE workflow people actually run: fit every model class, keep the AIC/BIC winner.

$\sim 10^4\times$

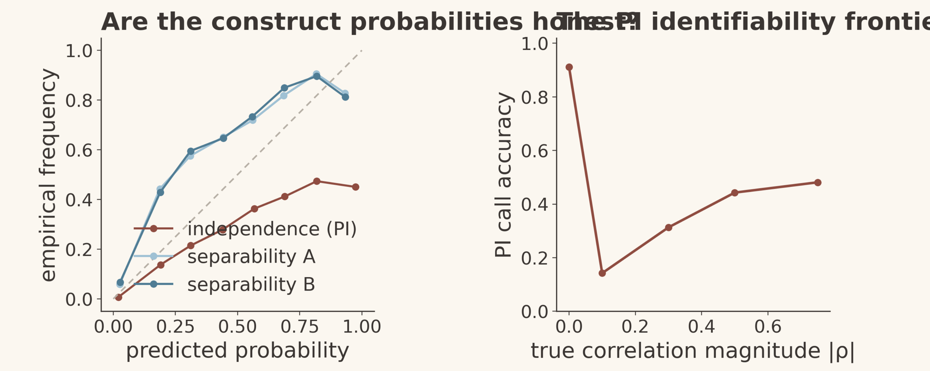
faster than the realistic MLE workflow

Fast enough is not a bonus here — it is the whole enabling condition. At ~3 µs a posterior update fits inside an inter-stimulus interval, so the model can sit *inside* the trial loop instead of waiting until the session is over.

And below ~100 trials per stimulus the network is also *more accurate* than MLE: the prior regularises exactly where likelihood optimisation diverges. It also never fails to return an answer, which the R implementations sometimes do.

LIMITS

Where does it break?



Construct probabilities are calibrated; the right-hand panel is the identifiability frontier for perceptual independence.

Separability is recovered sharply (≈ 0.94) and matches exact AIC/BIC selection. Perceptual independence does not: accuracy climbs with the size of the true correlation.

> **We are not overselling PI.** Near-independence is information-limited, so we report a calibrated *probability* rather than forcing a verdict. Decisional separability stays assumed — response times do not rescue it, and recovering it needs bound manipulations we have not run.

Adding response times helps where you would expect and not where you would hope: it buys processing architecture (0.88) and dimension neglect (0.97), but moves PI only $0.60 \to 0.65$.

THE PAYOFF

What does that buy you?

55%

of trials saved by stopping each participant at a precision threshold instead of a fixed budget

46%

fewer trials for a simulated learner to reach an expert target under a representation-tracking curriculum

An honest negative: information-gain stimulus selection roughly *ties* round-robin. With four symmetric stimuli, balanced coverage is already near-optimal. The win comes from knowing *when to stop* and *what to teach*, not from cleverer sampling.

> **Next.** Take this to melanoma screening, where border and colour genuinely overlap; relax decisional separability; and move from selecting stimuli in a pool to generating them. The targeted-training result rests on an *assumed* learning dynamic — it demonstrates the mechanism, not a validated protocol.

Ashby, F. G., & Townsend, J. T. (1986). Varieties of perceptual independence. *Psychological Review, 93*, 154–179.
Cranmer, K., Brehmer, J., & Louppe, G. (2020). The frontier of simulation-based inference. *PNAS, 117*, 30055–30062.
Silbert, N. H., & Thomas, R. D. (2013). Decisional separability, model identification, and statistical inference in GRT. *Psychonomic Bulletin & Review, 20*, 1–20.

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Every claim here is re-runnable: `python validation/run_all.py`